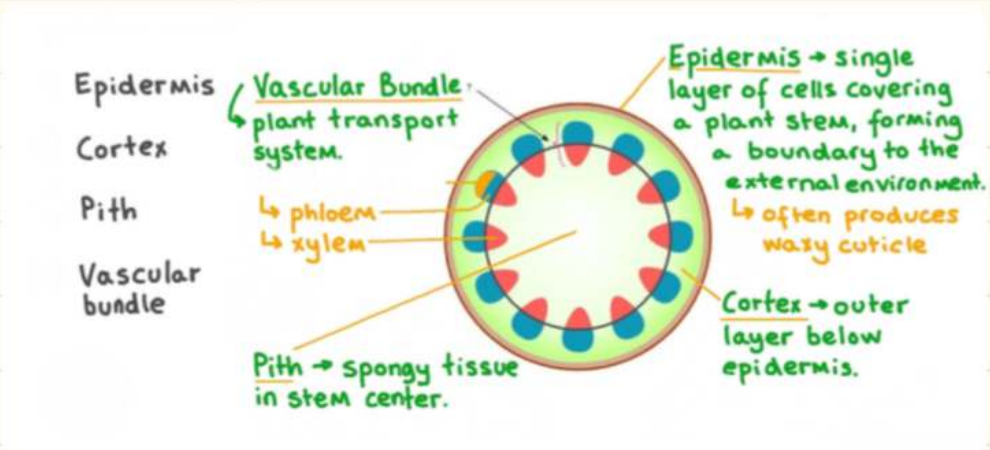
Transport in Plants

Functions of the xylem and phloem:

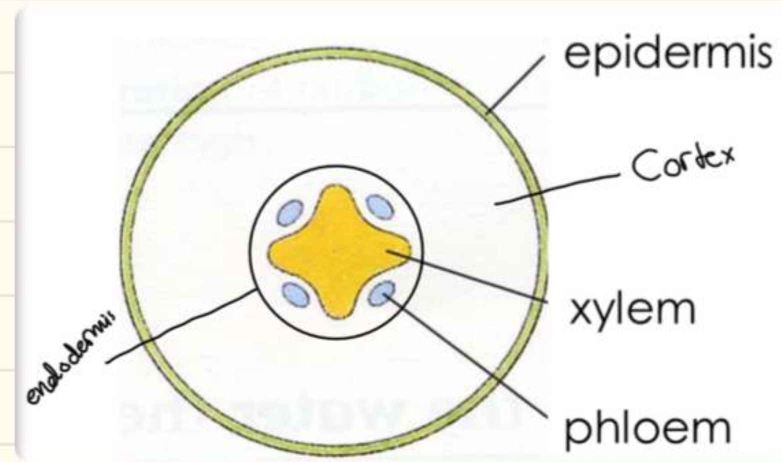
1- Xylem: Transport of water and mineral ions, and support

2- Phloem: Transport of sucrose and amino

In the stem



In the roots

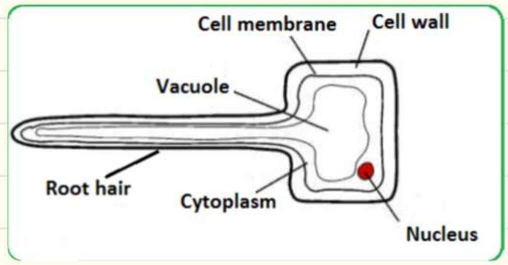


Adaptations of the xylem:

1- Thick walls with lignin, which makes the xylem rigid and waterproof.
2- No living contents
3- Elongated cells, and end of the cells are removed to form a long continuous tube

Roots

- Root tips are covered with root caps for protection as they grow into the soil.
- The rest of the roots are covered by epidermal cells from which root hairs extend. Root hair cells provide large surface area for absorption
- Water from the soil enters the roots by osmosis and moves from cell to cell until it gets to the xylem



Adaptations of the root hair cells

1- Have large surface area for absorption of water and minerals
2- Have many mitochondria to produce energy used in active transport of minerals
3- Are present in large numbers for maximum and efficient absorption

Water uptake in the xylem

- Water moves upwards in the xylem due to pressure difference as water in the roots have higher pressure than in leaves due to transpiration.

- Water moves up the xylem as a column due to the cohesion forces between the water molecules which makes them stick together.

- The adhesion forces allow the water molecules to stick to the walls of the xylem

- Water gets to the top of the xylem and moves to the mysophyll cells of the leaves by osmosis

Pathway of water

Root hair cells ⟶ root cortex cells ⟶ Xylem ⟶ mysophyll cells

Transpiration: The loss of water vapour from the leaves

* Water evaporates from the surface of the mysophyll cells into the air spaces and then diffuses out of the leaf through the stomat as water vapour

Transpiration stream is when water is constantly lost from the leaves due to transpiration, reducing the pressure at the top of the xylem. Which allows the water to be pulled up the stem from the roots to leaves

Water vapour loss is related to the large internal surface area provided by the interconnecting air spaces between mysophyll cells and the size and number of stomata.

Investigating the rate of transpiration in potted plant.

- A clear plastic bag is used since it is transparent and can allow light to pass through for photosynthesis to occur

- As a control to make sure the results of different environmental factors on the rate of transpiration, make sure to use the same species and age of the plant, with the same number and size of leaves, but cover the whole plant with a clear plastic bag to prevent water loss.

- For all the experiment plants cover the soil with a clear plastic bag to make sure no water is allowed to evaporate from the soil and the only water is lost due to transpiration.

Advantages of transpiration:

1- Allows water to move from roots to leaves (transpiration stream)

2- Helps the plant to cool down

Conditions affecting the rate of transpiration.

1- Temperature: As the temperature increases, the rate of transpiration increases as water evaporates quickly to cool the plant down, stomata is open

2- Humidity: As the humidity increases, the rate of transpiration decreases as the water potential is equal inside of the leaf and in the air, so no water vapour moves out of the leaf, stomata is closed.

3- Wind speed: As wind speed increases, the rate of transpiration increase, stomata is open

4- Water supply: As the water supply increases, the rate of transpiration, stomata is open

5- Light intensity: As light intensity increases, rate of transpsration increases as stomata is open are opened due to the plant doing photosynthesis

Wilting

When a leaf cell loses water, its turgor pressure will fall. This fall in pressure allows the water in the cell wall to enter the vacuole and so restore the turgor pressure. So, all the mysophyll cells will be losing water faster than they can absorb it from the xylem vessels, and so the leaf wilts. Water loss from the cell vacuoles results in the cells losing their turgor and becoming flaccid. A leaf with flaccid cells will be limp and the stem will droop.

Translocation: The movement of sucrose and amino acids from sources to sinks.

Sources: The parts of the plants that release sucrose or amino acids

Sink: The parts of the plants that use or store sucrose or amino acids.

If photosynthesis takes place during summer, where there is high light intensity, leaves will act as sources since they will be photosynthesising and producing glucose which will be converted to sucrose and translocated in the phloem to other parts of the plant like flower, stem, fruit, roots, and tubers, which will either convert sucrose to glucose to be used in respiration to release instant energy or convert it to startch and store it; these parts will act as sinks.

If photosynthesis doesn't occur; in winter where there is low light intensity, roots and tubers will act as source since the startch stored there will be converted to sucrose and translocated in the phloem to stems, buds, and leaves, where it will be converted to glucose to be used in respiration; these parts act as sinks.

How do guard cells
Control the opening & closing of
Stomata ?

1- When guard cells are turgid, they move further away from each other, since they swell from gaining lots of water, so the stomata opens.

2- When guard cells are flaccid, they move closer together since they get and the stomata closes.